



Nokia Asha 310

Nokia unveiled its Asha 310 dual-SIM smartphone. It provides the option to switch between two SIM's and a Wi-Fi connection



Red sues Sony

"Red" the professional camera maker has sued Sony for infringing two of its patents in Sony's new F-series pro camcorders

Digit 101

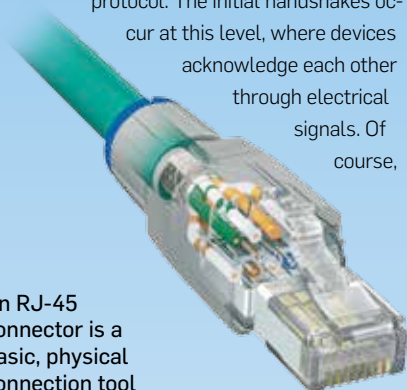
Internet Protocol 101

As you sit at your computer terminal, day in day out, making use of the Internet for a variety of things, have you ever spared a thought on how exactly the Internet works? How does one computer in rural India access information from a web page hosted on a server located in the United States in a split second? What rules govern the very fabric of how the Internet functions? It all happens through a well-established Internet protocol suite that operates in multiple layers. Let's take a closer look

Physical Link layer

This is the most rudimentary level of the multi-layered Internet protocol, and thereby also the most important one. Machine-to-machine connections through LAN cables and beyond are established at this layer of the Internet protocol. The initial handshakes occur at this level, where devices acknowledge each other through electrical signals. Of course,

An RJ-45 connector is a basic, physical connection tool



that's just putting it in a succinct way, for establishing connection between two machines isn't just about hooking them up through a piece of wire connected by an RJ-45 jack (or something else, for that matter).

Internet layer

If the physical link layer is the foundation stone of the Internet Protocol Suite, the Internet layer can be thought of as essentially forming the

beams and pillars of the robust framework sitting on top of it. Data packets that leave or enter a PC and their movement within any network – be it LAN or anything else – is governed by this layer. Identifying the appropriate gateway based on its IP address, establishing the right connections for transmitting and receiving data, and detecting errors in the chain and dealing with it is all the responsibility of the Internet layer. Not only identifying a destination machine through its IP address, but ensuring a data packet's security by encrypting information as it travels through the network is also handled by the Internet layer. At this point, the Internet can be visualized as a giant, sprawling spider web of devices interconnected with each other. But that's just it, there's nothing more to it – at least at this stage.

Transport layer

So far the two terminals are connected and have identified each other through their IP addresses, and about to begin the exchange of data. On top of these two layers sits the transport layer, which is primarily responsible for communication to take place. Transmission Control Protocol or TCP is a critical component of the Internet Protocol suite's data transport layer. Layers deployed above the transport layer relay information and communicate with other machines in the network through this layer.

Application layer

This is the layer above the transport layer, the topmost layer of the Internet Protocol suite, and one that's the easiest to understand and relate to. In fact, the Application layer and Transport layer are almost hand in glove, as they work very close to each other for all communication that takes place through the underlying network. Popular examples of application layer-related work that we do most of the times include FTP (file transfer protocol, useful for uploading files to a web host), HTTP (hyper text transfer protocol, or most of the readable Web we access through our browsers), and IRC (internet relay chat). The tools to send and receive emails, like



The Web is predominantly made of pages accessed over HTTP

SMTP, POP and IMAP all utilize the application layer of the IP suite. Even DNS or domain name servers, which is basically a naming system devised for systems connected to the Internet, also form part of the application layer. Not the easiest thing to wrap one's head around, the Internet Protocol suite, but that's the nature of the beast. 



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